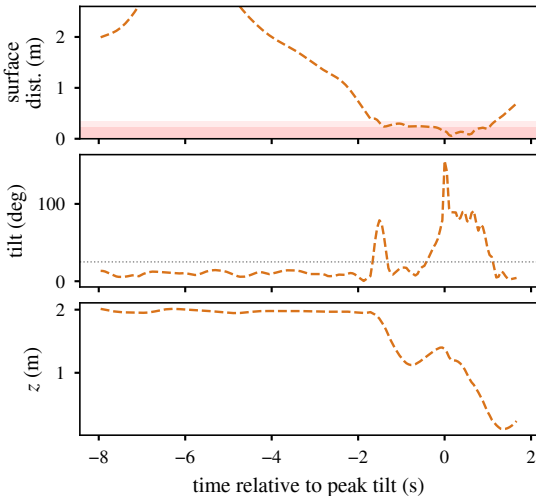
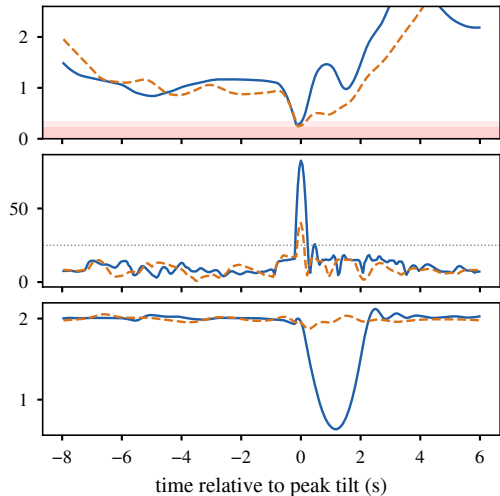


(a) contact leading to loss



(b) near-contact with recovery



--- rect, PID-LQR, iris_2 — plus, PID- H_∞ , iris_2 --- plus, PID-LQR, iris_2 ■ body contact ■ rotor contact possible